

CHIRANJEET DAS

SOFTWARE DEVELOPER

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SUMMARY

I am a Security and Software Engineer with a strong background in machine learning, AI, and deep learning. I specialize in building secure, scalable software systems, focusing on vulnerability analysis, encryption, and secure coding practices. I actively participate in Capture The Flag (CTF) challenges to sharpen my skills in ethical hacking and cybersecurity. Passionate about developing innovative solutions, I thrive in tackling complex technical challenges, optimizing workflows, and ensuring secure, robust infrastructures across various platforms. My goal is to apply my skills to drive impactful, secure software development and contribute to cutting-edge technological advancements.

PROJECTS (RECENT)

Hardware design and architecture (verilog)

1. 6502 processor core :: Designed the core of the 6502 processor .
2. Custom RISC based processor core :: Designed a modern custom processor core with a pipeline, execution unit, memory system, control unit .

System-Level Engineering

1. Custom Memory Allocator : Developed a custom memory allocator in C++ using multiple methods (freelist, basic, and buddy) to manage memory resources efficiently.
2. ChachaShell : Designed and built a custom shell in C++ for local file management, enabling directory navigation (cd), creation, deletion, and a recursive tree structure visualizer for displaying directory contents down to the leaf nodes. Also implemented file transfers from scratch.
3. ChachaGit : Built a custom version control system in C++ from scratch, utilizing a hybrid hashmap and linked list data structure. Integrated with ChachaShell to synchronize repository actions with the local file system.
4. Neural Network Implementation from Scratch : Constructed a fully functional neural network in C++, covering every aspect from forward propagation to backpropagation and loss calculation, achieving successful error reduction.
5. Shell automated Antimalware program : Constructed to scan files via generating Hash-values

Machine Learning and AI Development

1. Transformer Model for Text Summarization : Developed a Transformer model from scratch in Python (PyTorch) for abstractive text summarization, achieving coherent and contextually relevant summaries.
2. LLM Chatbot with Streamlit Interface : Created a large language model (LLM) chatbot in Python (PyTorch) from scratch, complete with a user interface built using Streamlit to enable interactive conversation.
3. Custom GAN and Security Bot with Siamese Network : Designed custom GAN architectures and pipelines in Python and implemented a Siamese network-based security bot capable of photo identification for secure access control.
4. Pix2Pix Model for Image-to-Image Translation : Implemented a Pix2Pix model from scratch using PyTorch for image-to-image translation tasks, including converting input images to enhanced output formats.
5. Satellite Image Mapping System : Developed a system that takes satellite images as input and generates accurate maps using deep learning techniques in PyTorch, enhancing the interpretation and analysis of satellite data.
6. Custom Cuda kernels : Developed multiple cuda kernels using cuda language for enhancing particular operation or networks like Convolutions and ray tracing.

PROFESSIONAL EXPERIENCE

Machine Learning (NLP) Intern - Government Project

1.Pointer-Generator Networks Implementation

- Developed pointer-generator networks from scratch for abstractive text summarization, combining extraction and generation methods for more accurate and fluent summaries.
- Enhanced summarization quality by integrating attention mechanisms and pointer networks to improve coverage of the most relevant content in long-form documents.

2.Text Summarization Development

- Contributed to the design and development of advanced text summarization models, focusing on improved abstraction techniques for better content interpretation and summarization.

3. Documentation

- Led the documentation efforts, creating clear and detailed documentation for the model's architecture, training process, and deployment, ensuring the project was well-documented for future updates and reproducibility.

AI/ML Lead at Small Town Talks

1.End-to-End MVP Development

- Spearheaded the development of end-to-end AI/ML MVPs, from conceptualization and model building (including bots for data scraping) to full deployment on EC2 instances, ensuring smooth scaling and operation in the cloud.

2.Bot Development and Data Scraping

- Developed and implemented intelligent bots for automated data scraping, optimizing data collection and processing pipelines, enabling the team to gather and utilize critical data efficiently.

3.Deployment on AWS EC2

- Managed the deployment of AI models and bots to EC2 instances, ensuring the smooth functioning of machine learning applications in the cloud, optimizing for scalability and operational performance.

4.Transformer Architectures

- Designed and implemented multiple transformer-based architectures in various NLP tasks (e.g., text summarization, sentiment analysis) to improve model performance and efficiency.

5.AI/ML Research and Development

- Led ongoing research and development efforts to explore and implement cutting-edge AI/ML technologies, driving innovation within the company and improving the effectiveness of deployed systems.

TECH STACKS

Programming Languages: C++/C ,Verilog,Cuda,Python, ARMv7, Bash-SHELL

Machine Learning & AI Frameworks: PyTorch,Scikit-learn

Machine Learning & AI architectures: NLP, Generative Models (GANs, Pix2Pix), Computer Vision,Transformers

DeepLearning Cloud Computing & Deployment: AWS, Docker, Streamlit

Development Tools & Environment: Vim, Visual Studio, Visual Studio Code, Jupyter Notebooks |

Data Handling & Storage: SQL/NoSQL, HDF5/CSV | Networking & Scripting: Sockets (C++), Python Scripts | Additional Tools & Libraries: Flask, OpenCV, PIL

SPECIALISATIONS

AI/ML System Design
Natural Language Processing (NLP)
Deep Learning & Neural Networks
Custom Systems Development (C++)
Computer Vision
Model Optimization & Performance Tuning
Binary Exploitation
SUID Exploitation
ROP Exploitation

EDUCATION

National Institute of Technology ,Durgapur
B-TECH (2022-present)
Civil Engineering